



Introduction to NLP

AIAA 4051 Lecture 10

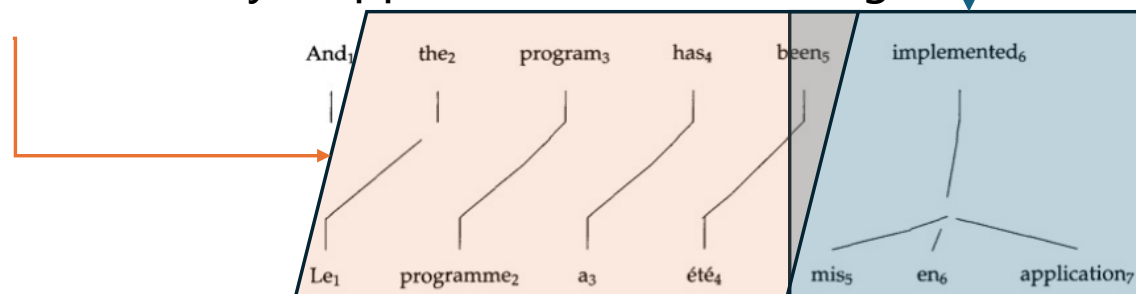
Instructor: Sihong Xie

Email: sihongxie@hkust-gz.edu.cn

AI Thrust, HKUST(GZ)

Weaknesses of Model 1

- IBM Model 1 makes several strong assumptions
 - Words f_1 and f_2 are generated independently $P(f_1|e_{a_1})$ and $P(f_2|e_{a_2})$.
 - all alignment matrices have the same probability.
- Both are not true
 - the last three French words are generated jointly from the last English word.
 - need the joint distribution $P(f_5, f_6, f_7|e_6)$ to model such linguistic dependencies;
 - locality: two consecutive French words are likely mapped to consecutive English words.
 - some alignments are unlikely.
 - can you give one?



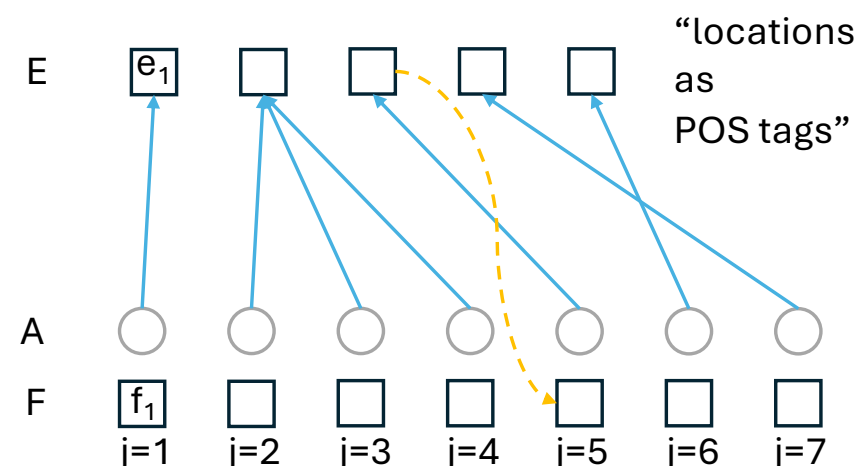
Adapt HMM for translation

- HMM alignment model (a generative model)
 - generates an alignment and the observed foreign sentence
 - predicting POS-tags => predicting alignment of a foreign word to a target location.
 - predicting observed words in a sentence => predicting a foreign word.
 - transition probabilities => align the next foreign word, given previous alignments.
 - emission probabilities => emit a foreign word given the up-to-date alignments.

Subscript: starting index
Superscript: ending index

$$\begin{aligned}
 P(f_1^J, a_1^J | e_1^I) &= P(J | e_1^I) \times \prod_{j=1}^J P(f_j, a_j | f_1^{j-1}, a_1^{j-1}, e_1^I) \\
 &= \boxed{P(J | e_1^I)} \times \prod_{j=1}^J \boxed{P(a_j | f_1^{j-1}, a_1^{j-1}, e_1^I)} \times \boxed{P(f_j | f_1^{j-1}, a_1^j, e_1^I)}
 \end{aligned}$$

Generate the length of the foreign sentence.



Adapt HMM for translation

- Make some Markov assumptions to simplify the above joint probability.

$$P(a_j | f_1^{j-1}, a_1^{j-1}, e_1^I) = P(a_j | a_{j-1}, I)$$

$$P(f_j | f_1^{j-1}, a_1^j, e_1^I) = P(f_j | e_{a_j})$$

- The final joint distribution ($a_j, j = 1, \dots, J$ are just random variables).

$$P(F, A | E) = P(J | I) \times \prod_{j=1}^J P(a_j | a_{j-1}, I) P(f_j | e_{a_j})$$

- How good an alignment is also depends on the generation of the foreign words.

- The translation model

$$P(F | E) = P(J | I) \times \sum_A \prod_{j=1}^J P(a_j | a_{j-1}, I) P(f_j | e_{a_j})$$

Adapt HMM for translation

- The transition probability should encourage alignment locality.
 - "the English words that generate neighboring Spanish words are likely to be nearby" – SLP 2nd Ed
 - $P(a_j | a_{j-1}, I)$ should be large if a_j is close to a_{j-1}
 - Locality is a relative concept and absolute positions are not relevant.
 - $P(7|6, I=15) = P(9|8, I=15)$.
 - Model "jumps" of the alignment pointers
 - $P(a_j | a_{j-1}, I)$ is a decreasing function of the jump $|a_j - a_{j-1}|$

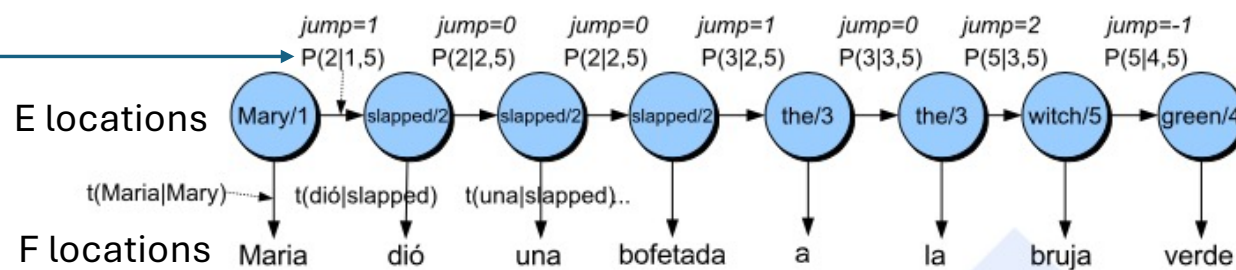
Example: $I=5$ for the translation.

Given:

$$a_1 = 1, a_2 = 2, a_3 = 2, a_4 = 2, \\ a_5 = 3, a_6 = 3, a_7 = 5,$$

$a_8 = ?$

(4 and 5 has higher probability than 1 or 2)



Decoding: generate a translation

- We have described two alignment models

- IBM Model-1

- HMM

$$P(F, A|E) = P(J|I) \times \prod_{j=1}^J P(a_j|a_{j-1}, I) P(f_j|e_{a_j})$$

- How to find the best translation E given a foreign sentence F and the model:

$$\hat{E} = \operatorname{argmax}_{E \in \text{English}} \underbrace{P(F|E)}_{\text{translation model}} \underbrace{P(E)}_{\text{language model}}$$

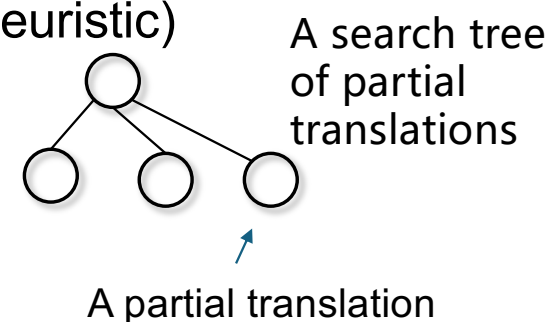
This is just a scoring function for evaluating translations.

- Note that E is unknown and it is different from aligning E and F.
- In general, decoding with a bigram language model is NP-complete (Knight 1999).
- HMM can use the tractable Viterbi algorithm.

Knight, K. (1999). Decoding complexity in word-replacement translation models. Computational Linguistics, 25(4), 607– 615.

Search-based methods

- In general translation, need “greedy” and “heuristic” search algorithms.
 - Believe that greedy is sufficient to find good quality translations (true for LLM).
 - Use heuristic cost function to guide the search (LLM is good heuristic)
- Partial search trees
 - state (node): partial translations



F

Maria	no	daba	una	bofetada	a	la	bruja	verde
Mary	not	give	a	slap	to	the	witch	green
	<u>did not</u>		<u>a slap</u>		<u>by</u>		<u>green witch</u>	
	<u>no</u>		<u>slap</u>		<u>to the</u>			
	<u>did not give</u>				<u>to</u>			
					<u>the</u>			
			<u>slap</u>			<u>the witch</u>		

- A partial translation:**
- An underline covers the scope of words in the sentence F .
 - Words above an underline are translation of words in the scope.

Search-based methods: best-first

- Best-first search
 - Always extend the node with the highest score.
 - The extensions are evaluated and en-queued.
 - Example ->
 - Expensive and short-sighted.

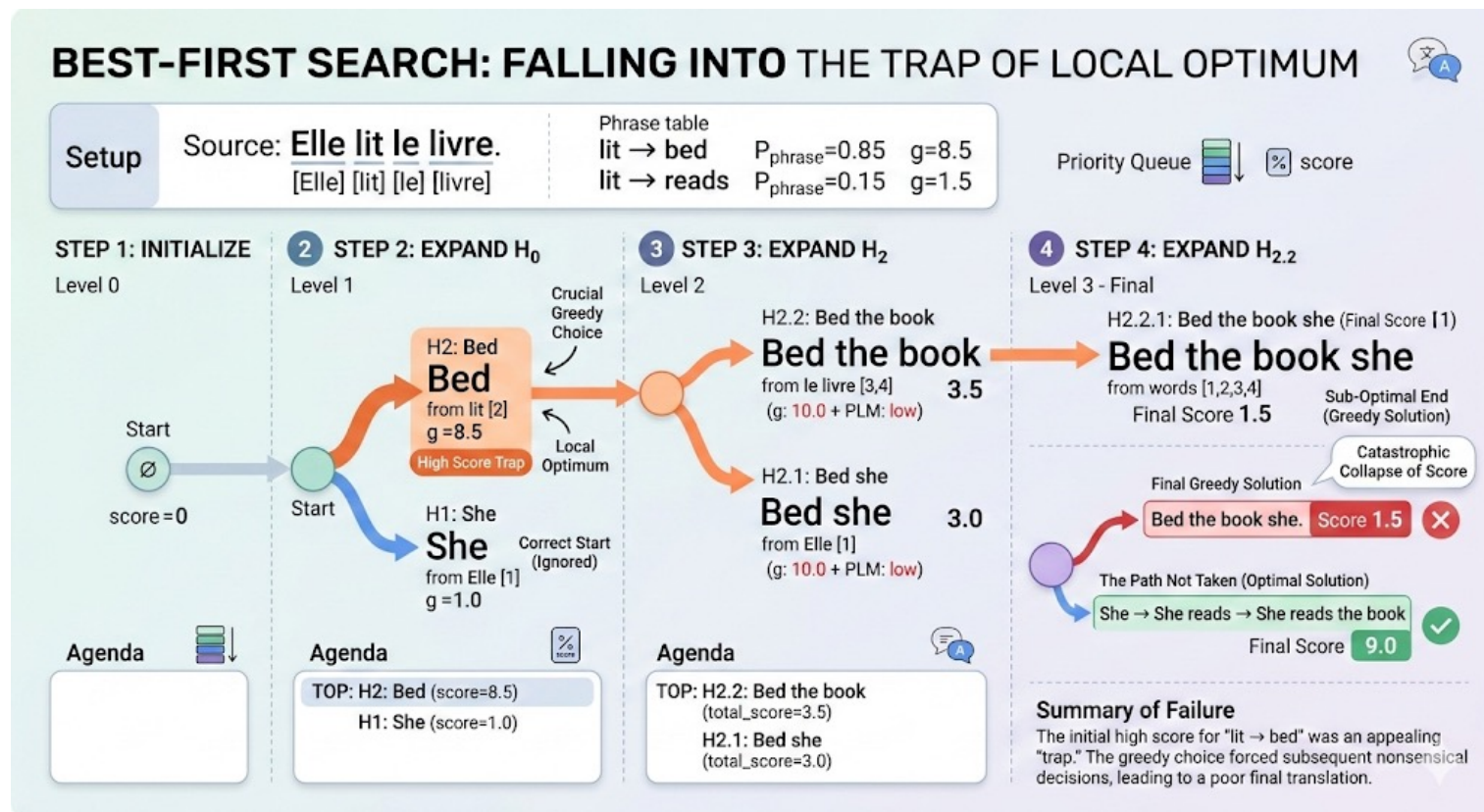


Image created by: Sihong with Gemini.

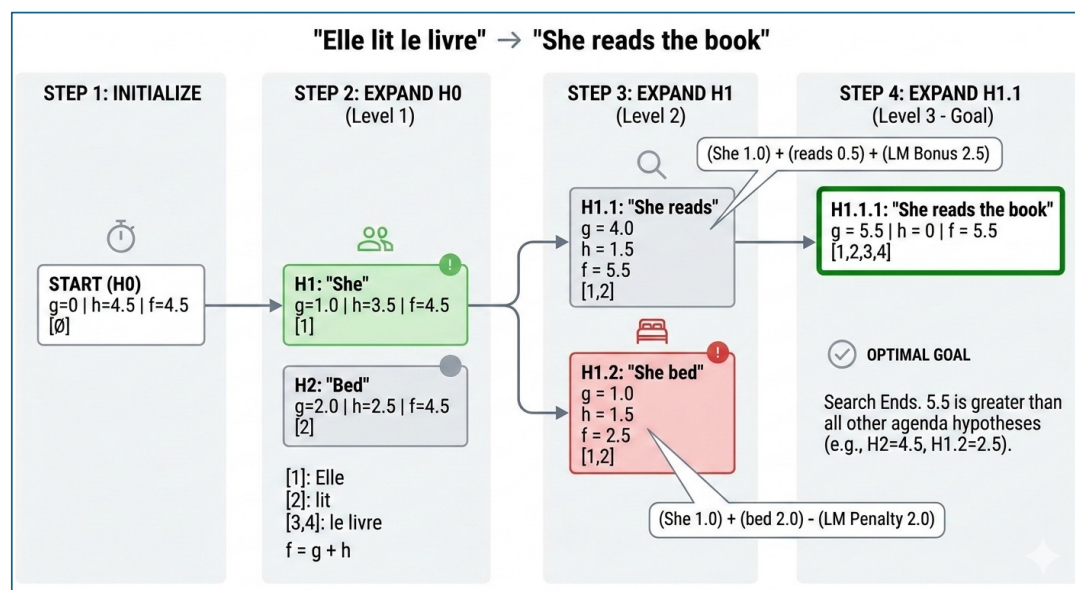
Search-based methods: A*

- A* search: better way to prioritize states for the next expansion.
 - Use a function to evaluate both **current** and **future** quality.

$$f^*(p) = g(p) + h^*(p)$$

The quality of the current partial translation

Translation quality for the **yet-to-translate** foreign words given p . Expensive to estimate and needs heuristics, which can be a simpler translation's probability of the remaining words using a phrase table.



Step 2: both hypotheses has the same score, and H1 is used since it covers the first word.

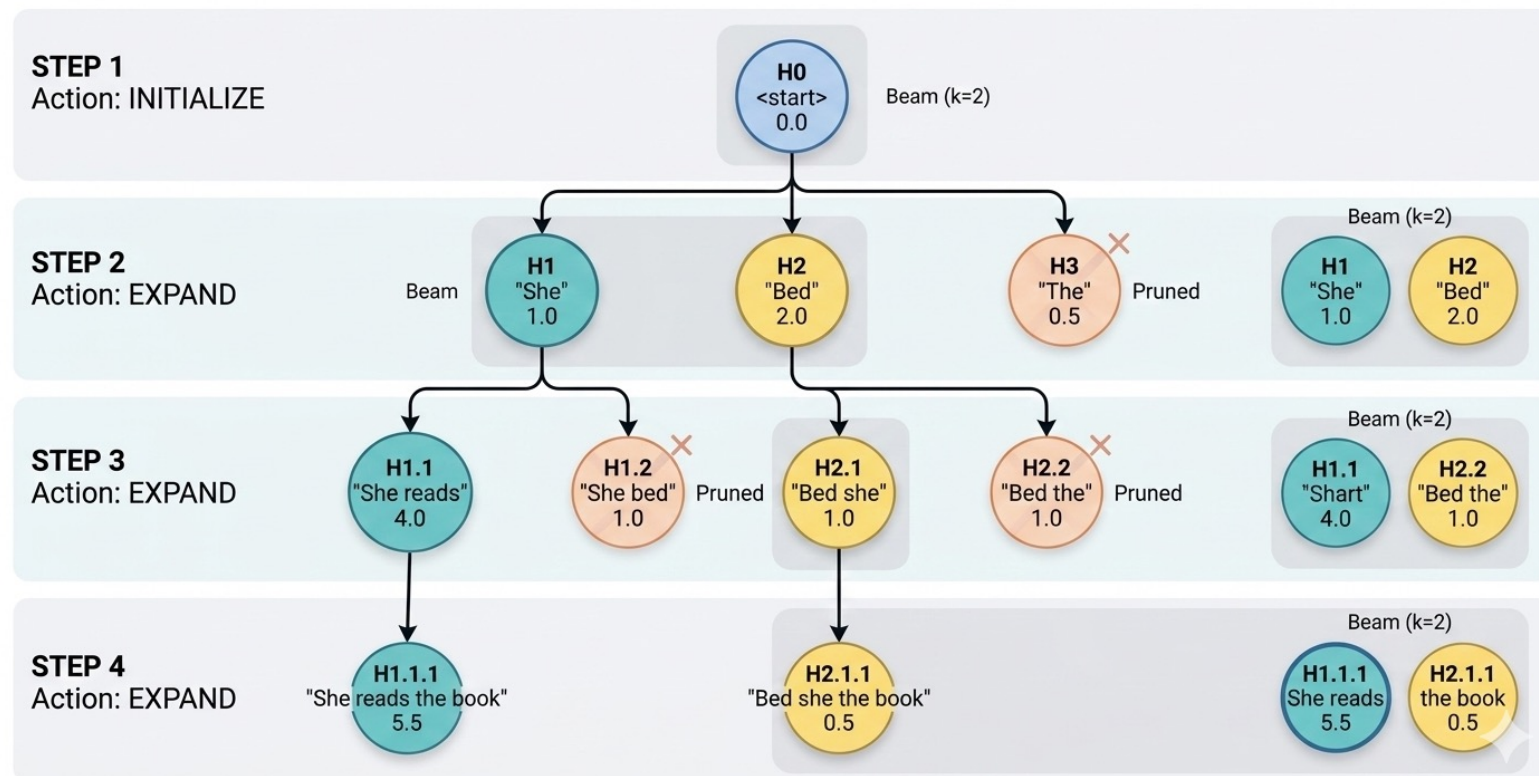
Step 3: $g(H1.1)=4$ using both translations of individual words and language model scores (LM), while the future scores are the same.

Step 4: as $f(H1.1.1)$ is greater than $f(H1.2)$ and $f(H2)$, the search ends.

Search-based methods: beam-search

- Beam search
 - Beam = a small set of the current k best states.
 - Expand all states in the beam but only keep the top k best expansions.

Simplistic Beam Search Trace Visualization ($k=2$)



Evaluating translations: BLEU score

- Automatic evaluation using BLEU score
 - Intuition: compute how often a predicted translation matches n-grams in any of the ground-truth translations.

Unigram precision:

17 hits/18 total words

Cand 1: It is a guide to action which ensures that the military always obeys the commands of the party

8 hits/14 total words

Cand 2: It is to insure the troops forever hearing the activity guidebook that party direct

Ref 1: It is a guide to action that ensures that the military will forever heed Party commands

Ref 2: It is the guiding principle which guarantees the military forces always being under the command of the Party

Ref 3: It is the practical guide for the army always to heed the directions of the party

BLEU: a method for automatic evaluation of machine translation, ACL, 2002

Caution of using BLEU: <https://slator.com/technology/how-bleu-measures-translation-and-why-it-matters/>

Image source: SLP, ed 2.

Evaluating translations: BLEU score

- Compute precision for n -grams, for $n=1,2,3,4$ for all candidate translations predicted by the model.

$$\text{prec}_n = \frac{\sum_{c \in \{\text{Candidates}\}} \sum_{n\text{-gram} \in C} \text{Count}_{\text{match}}(n\text{-gram})}{\sum_{c' \in \{\text{Candidates}\}} \sum_{n\text{-gram}' \in C'} \text{Count}(n\text{-gram}')}$$

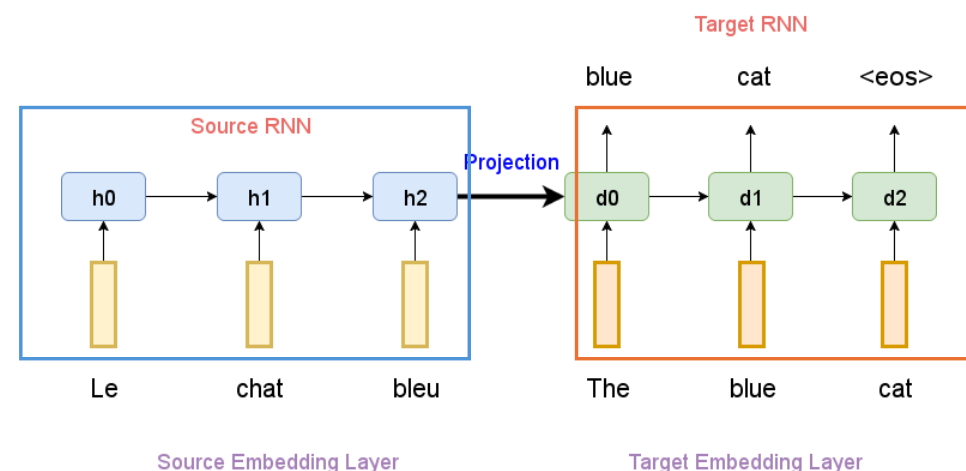
- Then take the geometric mean of these precisions $\left(\prod_{n=1}^4 \text{prec}_n\right)^{\frac{1}{4}}$ as the BLEU score of translating a sentence.
- This metric more comprehensively evaluate the fluency and faithfulness.

Evaluating translations: BLEU score

- Pitfalls in the evaluation using BLEU
 - Very short translations can have high precisions.
 - Example: "*for the*" will have precision 1 when the references contains the "*for the*", regardless of the remaining words.
 - Repetitive matches
 - Translation: "the the the the the the"
 - Reference: "the cat is on the mat"
 - Modified precision should be just 2/7 (2 = max number of "*the*" in the reference)

Seq2seq (encoder-decoder) model

- One RNN to encode the source sentence and one to generate a target sentence
 - **Encoder**: mapping from $\mathbf{x}$ (the input) to $\mathbf{h}$ (hidden units).
 - **Decoder**: mapping from the final $\mathbf{h}$ to $\mathbf{y}$ (the output).
- Input and output can have different lengths (good for machine translation, image captioning, music generation, etc.)
- Decouple the **encoder** and the **decoder**.

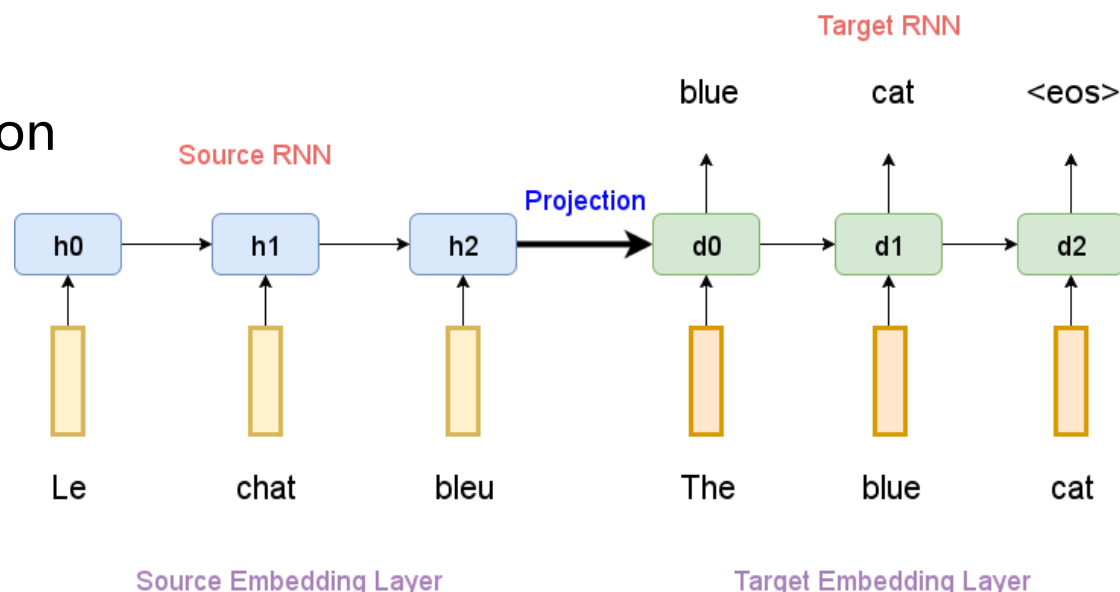


Sequence to Sequence Learning with Neural Networks, NIPS, 2015 (using LSTM)

Learning phrase representations using RNN encoder-decoder for statistical machine translation, 2015 (using RNN)

Seq2seq (encoder-decoder) model

Machine translation



Encoder is a regular RNN (source RNN):

- $$h^{(t)} = f(h^{(t-1)}, x^{(t)}; \theta)$$

Decoder is another regular RNN (target RNN):

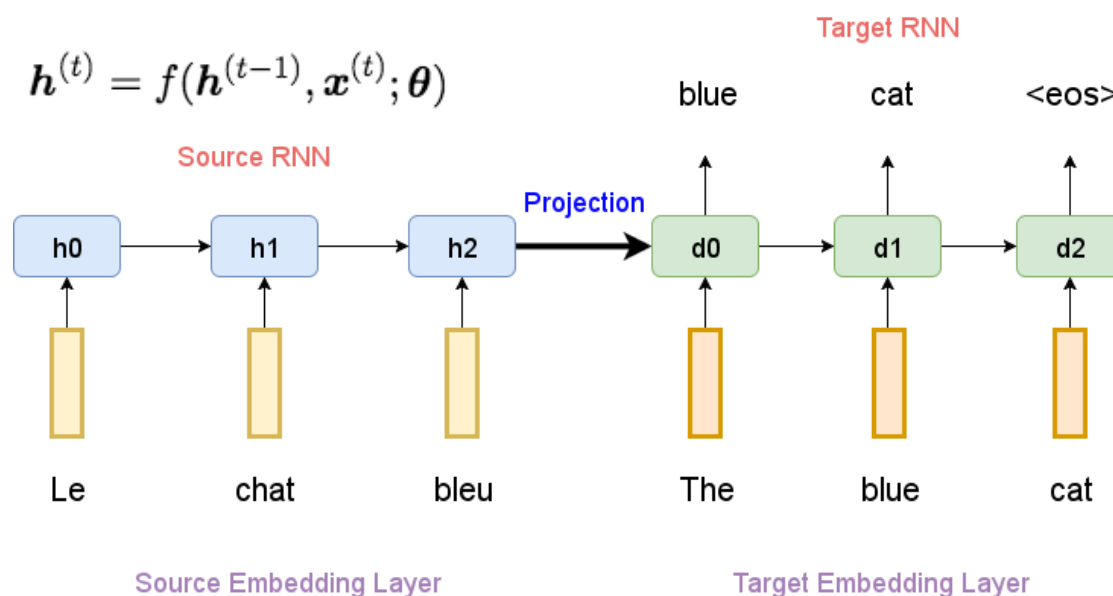
- $$d^{(t)} = g(d^{(t-1)}, \hat{y}^{(t-1)}; \theta)$$
- need the previous predicted word (called "**teacher forcing**") during test time.

source: <https://recordnotfound.com/Seq2Seq-PyTorch-MaximumEntropy-149015>

Seq2seq (encoder-decoder) model: training

Given an (input, output) pair $([x_1, \dots, x_T], [y_1, \dots, y_{T'}])$

$$\text{MLE: } p(y_1, \dots, y_{T'} | x_1, \dots, x_T) = \prod_{t=1}^{T'} p(y_t | v, y_1, \dots, y_{t-1})$$



$$p(y_t | v, y_1, \dots, y_{t-1}) = \text{softmax}(o^{(t)})$$

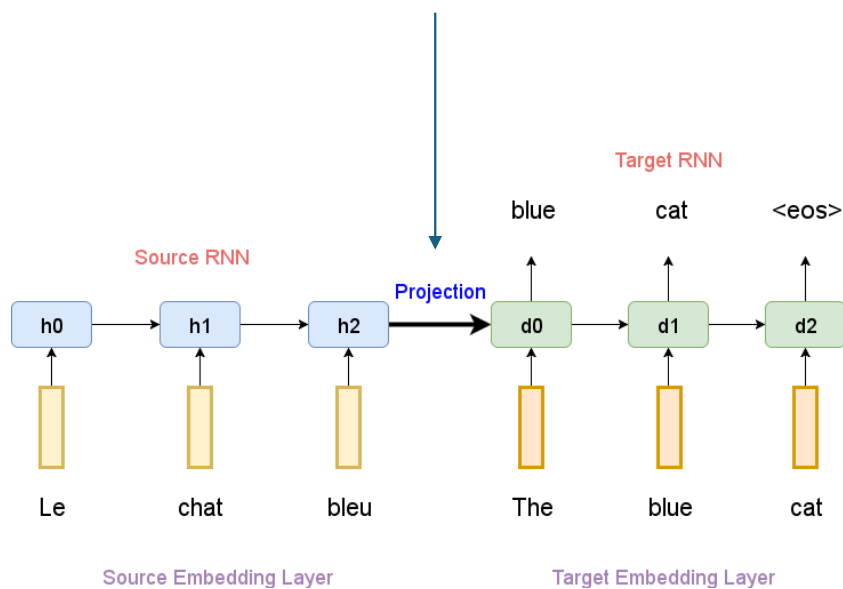
$$o^{(t)} = Vd^{(t)}$$

$$d^{(t)} = g(d^{(t-1)}, \hat{y}^{(t-1)}; \theta)$$

Decoder RNN calculating hidden states, using previously predicted output.

Seq2seq (encoder-decoder) model issues

1. The **fixed-length** hidden vector has to summarize all historic words and their ordering, regardless of the length of the source sequence. Asking for too much!

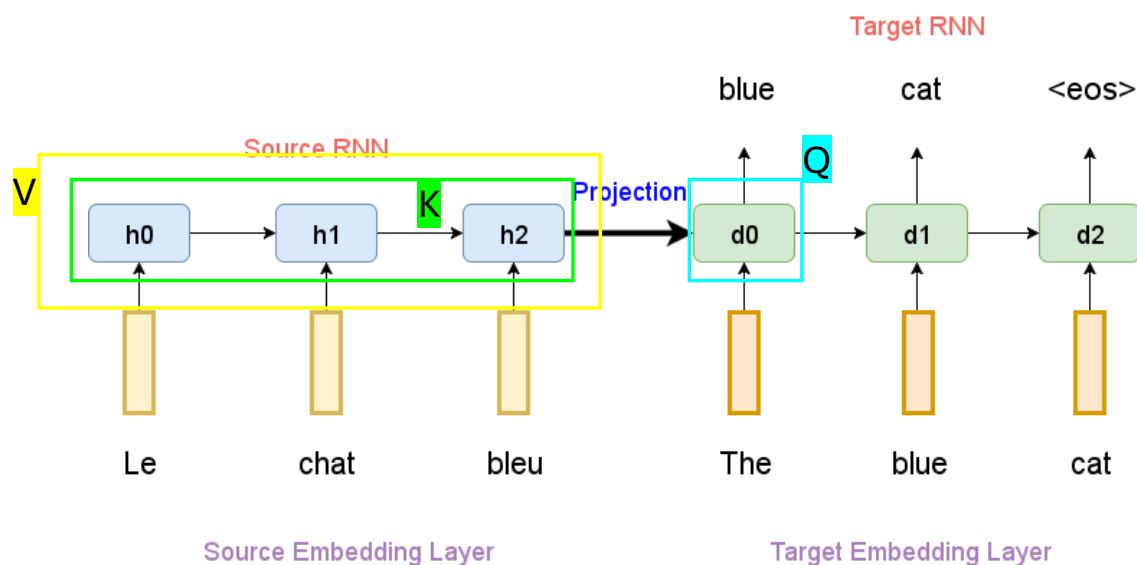


Seq2seq (encoder-decoder) model issues

- Fail to encode long-range dependencies:
 - E.g., *I am from China. I lived in Chicago for 6 years and then in Lehigh for another 6 years. I can speak fluent __?__*
 - Recent tokens' embeddings are less manipulated by matrix multiplications. As a result, RNN tends to have **sequential recency**
 - *The writer of the books are* (Incorrect)
 - *The writer of the books is* (Correct)
 - During backward propagation: error in the current token cannot be accurately reflected in the gradient w.r.t. much earlier tokens due to gradient explosion/vanishing (BPTT algorithm).
- Cannot be parallelized and can be slow for long sequences.

Attention mechanism

- What source positions are useful when generating “*cat*”?
- Learn to focus “end-to-end” rather than human specification:
 - let the training data drive the focus.



The idea is similar to search on Google:

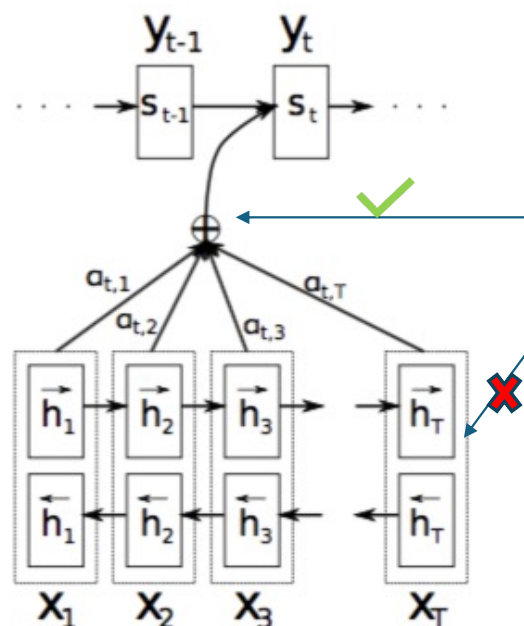
- Use a query **Q** to match important keywords (**K**) of webpages;
- Retrieve the webpage with the highest similarity.
- Extract information (the value **V**) from the webpage.

Neural machine translation by jointly learning to align and translate, ICLR, 2015

Attention mechanism

Decoder

Encoder
(bidirectional RNN)



Aggregate all relevant hidden states using attention weights instead of using the hidden states.

The query: s_{t-1}

The keys: $h_1 \dots h_T$

The value: $h_1 \dots h_T$

Hidden state of the decoder:

$$s_t = f(s_{t-1}, y_{t-1}, c_t)$$

Context c_t is the weighted sum:

$$c_t = \sum_{j=1}^T \alpha_{t,j} h_j$$

$$\alpha_{t,j} = \frac{\exp(\text{Score}(t, j))}{\sum_{k=1}^T \exp(\text{Score}(t, k))}$$

Example score functions

$$\text{Score}(t, k) = \langle s_{t-1}, h_t \rangle$$

$$\text{Score}(t, k) = s_{t-1}^\top W_a h_t$$

Advantage: can combine information from positions across the entire input.

Drawback: the encoder is still a sequential model and can be a computational bottleneck. Transformer will address this.

Neural machine translation by jointly learning to align and translate, ICLR, 2015

Extra readings

- For alignment and machine translation, see Chap 13 of FSNLP.
- For seq2seq, please find it in the paper
 - Sequence to Sequence Learning with Neural Networks, 2014 (Cited by 31310).
- For attention mechanism, please find it in the papers
 - Neural machine translation by jointly learning to align and translate, ICLR, 2015
 - Attention is all you need, NeurIPS 2017 (Cited by 235261)

| Demon

- Define a seq2seq model with GRU (replacing RNN) and attention in the decoder.
- Optimize the model on a toy example data.

Conclusion

- HMM can be used for machine translation. However, it is still a statistical model and cannot encode complex linguistic patterns.
- There are quite some search-based methods for decoding. It is hard to find global optimal (since there are only scoring functions and Viterbi algorithm requires stronger assumptions).
- RNN can be used to construct a larger model Seq2Seq.
- Attention mechanism is introduced to resolve the long-range dependencies issue in machine translation.

Quiz

1. Pen and paper; no compute or cellphone allowed.
 2. Turn in your answer sheet when you leave the classroom.
- Q1 (T/F): In IBM Model 1, two words are translated independently.
 - Q2 (T/F): By using HMM to model alignment matrix, a many-to-many translations can be implemented.
 - Q3 (short answer): if the target translation is "the tight jump to the hole" and a candidate translation is "the", calculate the uni-gram precision (BLEU-1 score) of this short candidate translation.
 - Q4 (multi-choice): what're the weaknesses of using seq2seq for translation: (A) cannot encode the entire input sentence; (B) earlier errors can lead to future mistakes; (C) lack of non-linear mapping in the model.
 - A: T, F, 1/1, (AB)